

DiffCSE-Inspired Sentence Embedding Learning using Contrastive Learning

Introduction

Sentence embeddings are dense vector representations that capture the semantic meaning of sentences. These embeddings are widely used in Natural Language Processing (NLP) tasks such as semantic similarity, information retrieval, clustering, and question answering.

Recent research has shown that contrastive learning can produce high-quality sentence embeddings without requiring manually labelled data. One such approach is DiffCSE (Difference-based Contrastive Learning for Sentence Embeddings) [1], which extends the SimCSE framework [2], by learning not only sentence similarity but also sentence differences through a discriminator-based objective.

In this project, a DiffCSE-inspired model was implemented using BERT and contrastive learning techniques. The objective was to learn meaningful sentence embeddings from unlabeled text and evaluate their quality on a semantic textual similarity benchmark.

The code is available at: [zahoor81/DiffCSE-Difference-based-ContrastiveLearning-for-Sentence-Embeddings](https://github.com/zahoor81/DiffCSE-Difference-based-ContrastiveLearning-for-Sentence-Embeddings)

Dataset

For training, the [Stanford Natural Language Inference \(SNLI\)](#) dataset was used.

Although SNLI contains labels such as: Entailment, Contradiction and Neutral

these labels were not used during training. Only the raw sentence text from the premise and hypothesis fields was extracted.

Example:

Premise: A man is playing football.

Hypothesis: A person is participating in a sport.

The labels associated with these sentence pairs were ignored, making the training process unsupervised. A total of approximately 20,000 sentences were selected for model training.

Model Architecture

Encoder

The encoder used in this project was:

bert-base-uncased

The model was loaded using the Hugging Face Transformers library.

The encoder converts each sentence into a contextual representation where each token receives a vector embedding.

Sentence Corruption

To create alternative views of the same sentence, random masking was applied.

Example:

Original sentence:

The cat is sitting on the mat.

Corrupted sentence:

The [MASK] is sitting on the [MASK].

This corruption process serves as a simple approximation of the editing process used in the original DiffCSE paper.

Mean Pooling

BERT produces embeddings for every token in a sentence.

To obtain a single sentence embedding, mean pooling was applied:

$$h = \frac{1}{N} \sum_{i=1}^N h_i$$

where:

- h_i = token embedding
- N = number of tokens

The resulting vector represents the entire sentence.

L2 Normalization

After mean pooling, embeddings were normalized using L2 normalization.

$$z = \frac{h}{\|h\|_2}$$

Normalization ensures that all embeddings have unit length and allows cosine similarity to focus on semantic direction rather than vector magnitude.

Contrastive Learning

The core learning mechanism used in this project was contrastive learning.

For each sentence:

1. Original sentence embedding was generated.

2. Corrupted sentence embedding was generated.
3. The model was trained to make embeddings from the same sentence more similar while pushing embeddings from different sentences apart.

This was achieved using the InfoNCE loss.

InfoNCE Loss

The similarity matrix was computed using:

$$S = Z_1 Z_2^T$$

where:

- Z_1 = original sentence embeddings
- Z_2 = corrupted sentence embeddings

Cross-entropy loss was then applied to maximize similarity between positive pairs and minimize similarity with negative pairs.

This enables the model to learn meaningful sentence representations without requiring labeled data.

Training Procedure

The model was trained using:

- Encoder: BERT Base Uncased
- Optimizer: AdamW
- Learning Rate: 7×10^{-6}
- Batch Size: 64
- Epochs: 2

The training loop consisted of:

1. Creating original and corrupted sentence pairs.
2. Encoding both sentences using BERT.
3. Computing InfoNCE loss.
4. Updating model parameters using backpropagation.

After training, the encoder weights were saved for future use.

Evaluation

The trained model was evaluated on the STS-Benchmark (STSB) dataset.

The [STS-B](#) dataset contains sentence pairs along with human similarity scores ranging from low similarity to high similarity.

For each pair:

1. Sentence embeddings were generated.
2. Cosine similarity was computed.
3. Predicted similarities were compared against human ratings.

Pearson Correlation

Pearson correlation measures the linear relationship between predicted scores and human scores.

$$r = \frac{cov(X, Y)}{\sigma_X \sigma_Y}$$

Higher values indicate stronger agreement with human judgments.

Spearman Correlation

Spearman correlation measures ranking consistency. It evaluates whether the model orders sentence similarities in the same way as humans. This metric is commonly reported in sentence embedding research papers.

Results

The proposed DiffCSE-inspired model was trained using the SNLI dataset in an unsupervised manner and evaluated on the STS-B test set.

Metric	Score
Pearson Correlation	0.6927
Spearman Correlation	0.6768

These results indicate that the learned sentence embeddings are able to capture semantic similarity reasonably well despite being trained without human-annotated similarity labels.

Comparison with Pretrained DiffCSE Model

To compare our implementation with an existing DiffCSE model, we loaded the publicly available pretrained model: "voidism/diffcse-bert-base-uncased-sts"

The same STS-B evaluation procedure was applied, yielding the following results:

Metric	Score
Pearson Correlation	0.6884
Spearman Correlation	0.6979

Conclusion

In this project, a DiffCSE-inspired sentence embedding model was implemented using BERT and contrastive learning. The model was trained in an unsupervised manner using raw sentences from the SNLI dataset without utilizing any labels. Sentence representations were learned through contrastive learning using original and corrupted sentence pairs. The resulting embeddings were evaluated on the STS-B benchmark using Pearson and Spearman correlation metrics.

The project demonstrates how self-supervised learning can produce useful sentence embeddings without requiring manually labeled training data and provides a practical introduction to modern sentence representation learning techniques.

References

- [1] Chuang, et al. (2022). DiffCSE: Difference-based Contrastive Learning for Sentence Embeddings. Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), 4207–4218.
- [2] Gao, T., Yao, X., & Chen, D. (2021). SimCSE: Simple Contrastive Learning of Sentence Embeddings. Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP), 6894–6910.